

GeoNatureAgent Benchmark: Benchmarking LLM Agents for Environmental Geospatial Analysis Across Frontier and Open-Weight Foundation Models [Benchmark]

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Abstract

Environmental scientists spend disproportionate effort on data wrangling rather than analysis, yet no benchmark exists to evaluate AI agents that automate environmental geospatial workflows through structured tool calling against real APIs. We introduce **GeoNatureAgent Benchmark**, a benchmark for environmental analysis agents comprising 93 tasks across 18 categories—including municipality-level analysis, multi-turn conversation, spatial reasoning, cross-indicator synthesis, error handling and recovery, ranking, comparison, multilingual understanding, habitat analysis, deep-dive profiling, temporal change, and task rejection—evaluated against an open, self-hostable geospatial API serving three environmental indicators across Spain and Portugal using twelve domain-specific tools. We evaluate seven large language models (Claude Sonnet 4, DeepSeek V3.2, GLM-5, Gemini 2.5 Pro, Qwen3-235B, GPT-OSS-120B, and Llama 4 Scout) across Vertex AI and Anthropic platforms under three temperature-1.0 seeds per model, reporting capability and per-case cost as orthogonal axes. We find that: (1) Claude Sonnet 4 leads on capability at $60.8\% \pm 0.8\%$, with DeepSeek V3.2 a close second at $56.3\% \pm 3.1\%$ and no other model exceeding 51%; (2) the cost-accuracy Pareto frontier is occupied entirely by open-weight models (Llama 4 Scout $\rightarrow$ Qwen3-235B $\rightarrow$ DeepSeek V3.2), with DeepSeek V3.2 at \$0.011/case offering 93% of Claude’s accuracy at 11 $\times$ lower cost; (3) comparison tasks remain universally unsolved (0% across all models on close-value comparisons), exposing systematic reasoning limitations; and (4) structured tool calling against a real API provides a more discriminative evaluation signal than synthetic benchmarks, with mean accuracies 25–35 percentage points below those reported on general-purpose GIS benchmarks. We demonstrate domain and geographic extensibility by incorporating BigEarthNet V2 land cover data for Portugal as a third indicator alongside Spanish CO₂ suitability and gully erosion, enabling cross-country evaluation across three data domains. GeoNatureAgent

Benchmark, the evaluation harness, and the self-hostable API are publicly available.¹

CCS Concepts

• **Computing methodologies** $\rightarrow$ **Neural networks**; *Multi-agent systems*; • **Information systems** $\rightarrow$ **Geographic information systems**; • **Applied computing** $\rightarrow$ *Earth and atmospheric sciences*.

Keywords

Geospatial AI, LLM Agents, Tool Calling, Benchmark, Environmental Analysis

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¹https://github.com/gabrielireland/GeoNatureAgent_Benchmark